

Olympiad Test Paper — Mathematics (Class 7) — Paper 3

Chapter 10: Operations with Integers

Paper 3 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	10 — Operations with Integers
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Watch the classic slips: subtracting a negative ADDS, the sign of a product is set by the parity (odd/even) of negative factors, multiplication and division bind tighter than + and - so brackets must be cleared in the right order, and a single hidden factor of 0 collapses the whole product. Do not write on this paper — mark answers on your answer sheet.

1. A token at 0 on a number line is moved +7, then -3, then doubled in distance from 0 keeping its sign, then -10. Where does it end up? (Move steps add; the doubling multiplies the current position by 2.)

- (A) +8 (B) +18
(C) -6 (D) -2

2. Compute $(-15) + (+8) + (-6) + (+11) + (-4)$.

- (A) -44 (B) +6
(C) -6 (D) -16

3. Which single statement is TRUE for the operation of subtraction on integers?

- (A) Subtracting any integer always makes the result smaller.
(B) Subtraction is not commutative: $a - b$ need not equal $b - a$.
(C) Subtraction is associative: $(a - b) - c$ always equals $a - (b - c)$.
(D) The integers are not closed under subtraction.

4. Evaluate $(-54) \div 9 \div (-2)$.

- (A) +3 (B) -3
(C) +27 (D) -27

5. Evaluate $(-4) + 3 \times (-5) - (-6)$.

(A) -41

(B) -13

(C) -25

(D) +5

6. Which expression equals 0 by the additive-inverse property without any arithmetic on magnitudes?

(A) $(-13) + 13$

(B) $(-13) \times 0$

(C) $(-13) + (-13)$

(D) $13 - (-13)$

7. For any integer a , the value of $a \times 1 \times (-1) \times 1$ is:

(A) a itself

(B) always negative

(C) the opposite of a

(D) always 0

8. At 6 a.m. the temperature was -9°C . It rose 4°C each hour for 3 hours. The temperature at 9 a.m. is:

(A) -21°C

(B) -5°C

(C) $+3^{\circ}\text{C}$

(D) $+12^{\circ}\text{C}$

9. Compute the product $(-1) \times (-2) \times (-3) \times (-4)$.

(A) -24

(B) +24

(C) -10

(D) +10

10. A diver descends from the surface (0 m) at 3 m every minute for 4 minutes, then rises 5 m. Her depth is:

(A) -17 m

(B) +7 m

(C) -12 m

(D) -7 m

11. Which sign-rule conclusion is INCORRECT?

(A) A negative integer times a negative integer gives a positive product.

(B) A negative integer divided by a negative integer gives a negative quotient.

(C) A positive integer minus a larger positive integer gives a negative result.

(D) A negative integer divided by a positive integer gives a negative quotient.

12. Evaluate $(-2) \times [7 - (-3)] + (-4) \times (-5)$.

(A) 0

(B) -40

(C) 40

(D) -20

13. The product of five non-zero integers is positive. Which is possible for the number of negative factors among them?

(A) 3

(B) 5

(C) 4

(D) 1

14. Evaluate $(-72) \div (-8) + (-3) \times (-2)$.

(A) +3

(B) +15

(C) -15

(D) +9

15. A bank account starts at ₹0. There are credits of ₹150 and ₹90 and debits (withdrawals) of ₹200 and ₹85. The balance is:

- (A) +₹45 (B) -₹45
(C) -₹525 (D) +₹525

16. By how much is (-7) greater than (-20) ?

- (A) -13 (B) 27
(C) 13 (D) -27

17. Which of these expressions has the GREATEST value?

- (A) $(-3) \times (-3) \times (-1)$ (B) $(-5) \times 2$
(C) $(-2) \times (-2) \times (-2) \times (-2)$ (D) $(-4) - (-1)$

18. Evaluate $20 - 36 \div (-4) + (-5)$.

- (A) +6 (B) +20
(C) -24 (D) +24

19. Which statement correctly describes the number 1 for integer multiplication?

- (A) 1 is the multiplicative identity: $a \times 1 = a$ for every integer a .
(B) 1 is the additive identity for integers.
(C) Multiplying by 1 always gives a positive result.
(D) 1 is the opposite of -1 under addition only.

20. Evaluate $(-1) \times (-2) \times (-3) \times (-4) \times (-5)$.

- (A) -120 (B) +120
(C) -15 (D) +15

21. A submarine at -300 m rises in equal steps and after 5 steps reaches -50 m. Each step was:

- (A) -50 m (B) +50 m
(C) +250 m (D) +70 m

22. For any integer n , $n \times (-1) \times (-1)$ equals:

- (A) $-n$ (B) always positive
(C) n (D) n^2

23. Compute $(-9) + (-9) + (-9) + (-9) + (-9) + (-9)$.

- (A) -45 (B) -54
(C) +54 (D) -63

24. Evaluate $(-18) \div [(-2) + (-1)] - (-4)$.

- (A) +2 (B) -10
(C) -2 (D) +10

25. Which property justifies rewriting $(-4) \times (7 + (-2))$ as $(-4) \times 7 + (-4) \times (-2)$?

- (A) associative property of multiplication (B) commutative property of addition
(C) additive identity
(D) distributive property of multiplication over addition

26. Evaluate $(-96) \div 8 \times (-1) \div (-4)$.

- (A) +3 (B) -3
(C) -12 (D) +12

27. In a test, +4 marks are given for each correct answer and -2 for each wrong one; unattempted score 0. Aman attempts 18 of 25 questions, getting 11 right and 7 wrong. His score is:

- (A) 30 (B) 44
(C) 58 (D) 16

28. Which value makes $(-6) \times \underline{\hspace{1cm}} = +30$ true?

- (A) +5 (B) -5
(C) -180 (D) +180

29. Evaluate $-5^2 + (-5)^2$.

- (A) 50 (B) -50
(C) 0 (D) -10

30. Evaluate $[(-8) - (-3)] \times [(-2) - 4]$.

- (A) +30 (B) -30
(C) +10 (D) -10

31. The sum of an integer and its opposite is then multiplied by (-17) . The result is:

- (A) -17 (B) +17
(C) -34 (D) 0

32. Which value of the box satisfies $\underline{\hspace{1cm}} \div (-6) = +7$?

- (A) +42 (B) -13
(C) +1 (D) -42

33. On a number line you start at +5, move 8 left, then 3 left, then 9 right, then 6 left. Your final position is:

- (A) -3 (B) +3
(C) -11 (D) -5

34. Evaluate $(-7) \times (-4) \times 0 \times (-9) + (-3) \times (-2)$.

- (A) 0 (B) +504
(C) -6 (D) +6

35. Which is the correct evaluation of $(-10) - 6 \times (-2) \div 3$?

(A) Apply $\times$ and $\div$ before the subtraction, left to right: $6 \times (-2) = -12$, then $-12 \div 3 = -4$, so $(-10) - (-4) = -6$.

(B) Work strictly left to right ignoring precedence: $(-10) - 6 = -16$, then $(-16) \times (-2) = 32$, $32 \div 3$ is about 10.7.

(C) $(-10) - 6 \times (-2) = (-10) + 12 = 2$ first, then $2 \div 3$ is about 0.7.

(D) $6 \times (-2) \div 3 = -4$, then read the minus as 'add': $(-10) + (-4) = -14$.

36. By how much does $(-5) \times 8$ differ from $5 \times (-8)$? (Give the size of the difference.)

(A) 80

(B) 0

(C) -80

(D) 16

37. Evaluate $(-3)^3$.

(A) +27

(B) -9

(C) -27

(D) +9

38. A shop's daily results were: loss ₹40, profit ₹130, loss ₹75, profit ₹25, loss ₹90. Over the five days the shop:

(A) gained ₹50

(B) lost ₹360

(C) lost ₹50

(D) gained ₹155

39. If p and q are both negative integers, then $p \times q \times (p + q)$ is:

(A) negative

(B) positive

(C) zero

(D) cannot be determined

40. Evaluate $(15 - 23) \times (-5) + (-40) \div 8$.

(A) 45

(B) -45

(C) 25

(D) 35

41. The product $(-1) \times (-2) \times (-3) \times \dots \times (-9)$ (all integers from -1 down to -9) is:

(A) positive

(B) zero

(C) -45

(D) negative

42. Which value makes the box true: $(-4) \times \underline{\hspace{1cm}} + 6 = -18$?

(A) 6

(B) -6

(C) 3

(D) -3

43. Evaluate $(-45) \div (-9) - (-2) \times 3$.

(A) -1

(B) 11

(C) -11

(D) 1

44. Evaluate $(-2) \times (-5) - (-3) \times (-4)$.

(A) +22

(B) +2

(C) -22

(D) -2

45. The night temperature was 2°C . It fell 5°C , then fell another 4°C , then rose 3°C . The temperature now is:

- (A) $+4^{\circ}\text{C}$
(C) -14°C

- (B) -4°C
(D) -8°C

46. Which expression equals the SMALLEST (most negative) integer?

(A) $(-5) \times (-5)$

(B) $(-6) + (-6)$

(C) $(-2) \times (-3) \times (-4)$

(D) $(-10) - (-1)$

47. Evaluate $(-64) \div (-8) \div (-2) - (-3)$.

(A) -1

(B) $+7$

(C) -7

(D) $+1$

48. Using a distributive step, $(-8) \times 102$ is best written as:

(A) $(-8) \times 100 + 8 \times 2 = -784$

(B) $(-8) \times 100 - (-8) \times 2 = -784$

(C) $(-8) \times 100 + (+16) = -784$

(D) $(-8) \times 100 + (-8) \times 2 = -816$

49. The product of two integers is $+48$ and one of them is -6 . The other integer is:

(A) $+8$

(B) -8

(C) -288

(D) $+288$

50. A delivery rider's earnings change by $-\text{₹}30$ (fuel), $+\text{₹}120$ (fares), $-\text{₹}45$ (fine), and $+\text{₹}80$ (tip) in a shift. Twice this net change equals:

(A) $+\text{₹}125$

(B) $+\text{₹}250$

(C) $-\text{₹}250$

(D) $+\text{₹}550$

51. Which list of integers is in DECREASING (descending) order?

(A) $6, 1, 0, -4, -9$

(B) $-9, -4, 0, 1, 6$

(C) $6, 0, 1, -9, -4$

(D) $-4, -9, 0, 1, 6$

52. Evaluate $(-6) \times [(-2) - (-5)] - (-6) \times (-2)$.

(A) -6

(B) $+30$

(C) -30

(D) $+6$

53. The integers are closed under which set of operations?

(A) addition, subtraction, and multiplication

(B) addition, subtraction, multiplication, and division

(C) addition and multiplication only

(D) subtraction and division only

54. Evaluate $(-2)^3 + (-3)^2 - (-1)^5$.

(A) 0

(B) -2

(C) -16

(D) $+2$

55. Two integers have a sum of -1 and a product of -30 . The integers are:

(A) -5 and 6

(B) -10 and 3

(C) -15 and 2

(D) 5 and -6

56. Evaluate $(-9) - 5 \times (-4) + (-12) \div 6$.

(A) -9

(B) $+13$

(C) $+9$

(D) $+11$

57. Start at floor $+2$ in a building with basements numbered negatively. The lift goes down 7, up 3, down 5, then up 6 floors. It ends at floor:

(A) $+1$

(B) -3

(C) -1

(D) $+5$

58. Which equals $(-14) - (-6) - (-9) - 4$?

(A) -33

(B) $+3$

(C) -3

(D) -9

59. The product of an ODD number of negative integers (all non-zero) is always:

(A) negative

(B) positive

(C) zero

(D) equal to the count of negatives

60. Evaluate $[(-5) \times (-2) - 4] \div [(-1) + (-2)]$.

(A) $+2$

(B) -6

(C) $+6$

(D) -2

Solutions — Mathematics (Class 7), Chapter 10: Operations with Integers — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	B	31	D	41	D	51	A
2	C	12	A	22	C	32	D	42	A	52	C
3	B	13	C	23	B	33	A	43	B	53	A
4	A	14	B	24	D	34	D	44	D	54	D
5	B	15	B	25	D	35	A	45	B	55	D
6	A	16	C	26	B	36	B	46	C	56	C
7	C	17	C	27	A	37	C	47	A	57	C
8	C	18	D	28	B	38	C	48	D	58	C
9	B	19	A	29	C	39	A	49	B	59	A
10	D	20	A	30	A	40	D	50	B	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Start 0, $+7 \rightarrow 7$, $-3 \rightarrow 4$. Doubling the position: $4 \times 2 = 8$. Then $-10 \rightarrow 8 - 10 = -2$. Forgetting the -10 leaves 8; doubling before the -3 would give a wrong larger value; treating the final step as $+10$ gives 18.

2. (C) Group positives $8 + 11 = 19$ and negatives $(-15) + (-6) + (-4) = -25$. Total $19 - 25 = -6$. Adding all magnitudes as if negative gives -44 ; the opposite sign mistake gives $+6$.

3. (B) $a - b$ and $b - a$ differ in sign in general, so subtraction is not commutative. Subtracting a negative increases the result, so it does not always shrink. $(a - b) - c \neq a - (b - c)$ in general, so it is not associative. Integers ARE closed under subtraction, so that option is false.

4. (A) Left to right: $(-54) \div 9 = -6$, then $(-6) \div (-2) = +3$ (negative $\div$ negative is positive). Stopping the sign at -3 ignores the second negative; multiplying instead of dividing the 2 gives ± 27 .

5. (B) Multiply first: $3 \times (-5) = -15$. Then $(-4) + (-15) - (-6) = -4 - 15 + 6 = -13$. Adding -4 and 3 first (wrong order) gives $(-1) \times (-5) \dots =$ different value; treating $-(-6)$ as -6 gives -25 .

6. (A) An integer plus its opposite is 0 (additive inverse). $(-13) + 13 = 0$ by that property. $(-13) \times 0$ is 0 but by the zero-product rule, not additive inverse; $(-13) + (-13) = -26$; $13 - (-13) = 26$.

7. (C) Multiplying by 1 leaves a unchanged; multiplying by (-1) gives the opposite. So $a \times 1 \times (-1) \times 1 = -a$. It is not a itself (one negative factor flips the sign), not always negative (if a is negative, $-a$ is positive), and not 0.

8. (C) Rise over 3 hours $= 3 \times 4 = 12$ °C. So $-9 + 12 = +3$ °C. Subtracting 12 gives -21 ; counting only one hour gives -5 ; forgetting the start of -9 gives $+12$.

9. (B) Four negative factors $\rightarrow$ even count $\rightarrow$ product positive. Magnitude $1 \times 2 \times 3 \times 4 = 24$, so $+24$. An odd-count error gives -24 ; adding instead of multiplying gives ± 10 .

10. (D) Descent: $4 \times 3 = 12$ m down $\rightarrow -12$ m. Rising 5 m: $-12 + 5 = -7$ m. Subtracting the rise gives -17 ; ignoring the descent rate gives -12 ; a sign flip gives $+7$.

11. (B) Negative $\div$ negative is POSITIVE, so that statement is the false one. The other three are all correct sign facts: $(-)(-)=+$, a smaller minus a larger is negative, and $(-)\div(+)$ is negative.

12. (A) Inside the bracket: $7 - (-3) = 10$. Then $(-2) \times 10 = -20$, and $(-4) \times (-5) = +20$. Sum: $-20 + 20 = 0$. Treating $7 - (-3)$ as 4 changes the first term; making both products negative gives -40 .

13. (C) A positive product needs an EVEN number of negative factors. Among 0–5, the even possibilities are 0, 2, 4. Of the listed options only 4 is even. 3, 5, and 1 are odd and would give a negative product.

14. (B) Division: $(-72) \div (-8) = +9$. Multiplication: $(-3) \times (-2) = +6$. Add: $9 + 6 = 15$. Subtracting the 6 gives $+3$; making the division negative gives -15 .

15. (B) Credits add: $150 + 90 = 240$. Debits subtract: $200 + 85 = 285$. Balance $240 - 285 = -45$, i.e. overdrawn ₹45. A sign slip gives $+45$; adding everything as same sign gives 525.

16. (C) 'How much greater' means $(-7) - (-20) = -7 + 20 = 13$. Treating the subtraction of a negative as a subtraction gives -27 ; a sign slip gives -13 .

17. (C) $(-2)^4 = +16$ (four negatives, positive). $(-3) \times (-3) \times (-1) = -9$; $(-5) \times 2 = -10$; $(-4) - (-1) = -3$. The largest is $+16$.

- 18. (D)** Division first: $36 \div (-4) = -9$, so the expression is $20 - (-9) + (-5) = 20 + 9 - 5 = 24$. Doing $20 - 36$ first breaks order of operations; treating $-(-9)$ as -9 gives $+6$.
- 19. (A)** $a \times 1 = a$ defines 1 as the multiplicative identity. 0 (not 1) is the additive identity. Multiplying by 1 keeps the original sign, so it is not always positive. 1 and -1 are opposites, but the identity statement is the one being asked.
- 20. (A)** Five negative factors $\rightarrow$ odd $\rightarrow$ product negative. Magnitude $1 \times 2 \times 3 \times 4 \times 5 = 120$, so -120 . An even-count error gives $+120$; adding gives ± 15 .
- 21. (B)** Total rise = $(-50) - (-300) = 250$ m up over 5 steps $\rightarrow 250 \div 5 = +50$ m each. The $+/-$ sign of rise must be positive (it went up); -50 reverses direction; 250 is the whole rise, not one step.
- 22. (C)** $(-1) \times (-1) = +1$, so $n \times 1 = n$. The two negatives cancel; a single-negative slip gives $-n$; it is not n^2 (no multiplication of n by itself).
- 23. (B)** Six copies of -9 : $6 \times (-9) = -54$. Five copies would give -45 ; seven give -63 ; a sign slip gives $+54$.
- 24. (D)** Bracket: $(-2) + (-1) = -3$. Then $(-18) \div (-3) = +6$. Finally $6 - (-4) = 6 + 4 = 10$. Treating $6 - (-4)$ as $6 - 4$ gives $+2$; making the division negative gives -10 .
- 25. (D)** Multiplying a factor across a sum, $a \times (b + c) = a \times b + a \times c$, is the distributive property. Associativity regroups one operation; commutativity reorders; identity involves 0 — none of these matches the split shown.
- 26. (B)** Left to right: $(-96) \div 8 = -12$; $\times (-1) = +12$; $\div (-4) = -3$. Two of the three sign-flips net... counting the negatives: factors $-96, -1, -4 \rightarrow$ three negatives $\rightarrow$ odd $\rightarrow$ result negative, magnitude $96 \div 8 \div 4 = 3$, so -3 . An even-count error gives $+3$.
- 27. (A)** Correct: $11 \times 4 = 44$. Wrong: $7 \times (-2) = -14$. Score $44 - 14 = 30$; the 7 unattempted add 0. Ignoring penalties gives 44; counting all 25 as attempted changes the wrong count; adding the penalty as positive gives 58.
- 28. (B)** We need $(-6) \times x = +30$. Since the product is positive and one factor is negative, x must be negative: $x = 30 \div (-6) = -5$. Choosing $+5$ gives -30 ; multiplying instead of dividing gives ± 180 .
- 29. (C)** -5^2 means square first then negate: $-(25) = -25$. $(-5)^2 = +25$. Sum: $-25 + 25 = 0$. Treating -5^2 as $(-5)^2$ gives 50; sign slips give -50 or -10 .
- 30. (A)** First bracket: $(-8) - (-3) = -8 + 3 = -5$. Second: $(-2) - 4 = -6$. Product $(-5) \times (-6) = +30$ (two negatives). Treating the first bracket as -11 or making the product negative gives the wrong options.

31. (D) An integer plus its opposite is 0 (additive inverse), and $0 \times (-17) = 0$ (zero-product rule). Confusing the inverse sum with 1 gives -17 ; doubling gives -34 .

32. (D) If $x \div (-6) = +7$ then $x = 7 \times (-6) = -42$. Check: $(-42) \div (-6) = +7$. Choosing $+42$ gives -7 ; adding instead of multiplying gives -13 .

33. (A) Net move $= -8 - 3 + 9 - 6 = -8$. Start $+5$: $5 - 8 = -3$. Adding the rights and lefts with wrong signs gives the other options; $+5 - 8 = -3$ is correct.

34. (D) The first product contains a factor 0, so it is 0. The second product: $(-3) \times (-2) = +6$. Total $0 + 6 = 6$. Forgetting the second term gives 0; ignoring the 0 in the first term gives a large value; a sign slip gives -6 .

35. (A) Multiplication and division bind tighter than subtraction and are done left to right: $6 \times (-2) = -12$, then $-12 \div 3 = -4$. Finally $(-10) - (-4) = -10 + 4 = -6$. Subtracting first, or multiplying the two leftmost terms, or turning the final subtraction into an addition of a negative, each breaks the order-of-operations rule and gives a different (wrong) value.

36. (B) $(-5) \times 8 = -40$ and $5 \times (-8) = -40$. They are equal, so the difference is 0. Adding the two magnitudes gives 80; a sign confusion gives -80 .

37. (C) $(-3)^3 = (-3) \times (-3) \times (-3)$. First $(-3) \times (-3) = +9$, then $+9 \times (-3) = -27$ (three negatives $\rightarrow$ odd $\rightarrow$ negative). $+27$ forgets the odd parity; ± 9 stops after two factors.

38. (C) Profits: $130 + 25 = 155$. Losses: $40 + 75 + 90 = 205$. Net $155 - 205 = -50$, a loss of ₹50. A sign slip gives a gain of 50; summing all magnitudes as loss gives 360; ignoring losses gives 155.

39. (A) $p \times q$ is positive (two negatives). $p + q$ is the sum of two negatives, hence negative. Positive $\times$ negative = negative. It cannot be zero since p, q are non-zero negatives, and the sign is fully determined.

40. (D) Brackets: $15 - 23 = -8$, so $(-8) \times (-5) = +40$. Division: $(-40) \div 8 = -5$. Sum $40 + (-5) = 35$. Treating the division as $+5$ gives 45; making the product negative gives -45 .

41. (D) There are nine negative factors; nine is odd, so the product is negative. (None of the factors is 0, so it is not zero.) Mistaking nine for an even count gives positive; -45 is the sum, not the product.

42. (A) $(-4) \times x + 6 = -18 \rightarrow (-4) \times x = -24 \rightarrow x = (-24) \div (-4) = +6$. Check: $(-4) \times 6 = -24$, $-24 + 6 = -18$. Choosing -6 gives $+24 + 6 = 30$; sign/division slips give ± 3 .

43. (B) Division: $(-45) \div (-9) = +5$. Multiplication: $(-2) \times 3 = -6$, and there is a minus before it: $-(-6) = +6$. So $5 + 6 = 11$. Treating $-(-2) \times 3$ as -6 gives -1 ; a sign slip on the division gives -11 .

- 44. (D)** First product: $(-2) \times (-5) = +10$. Second product: $(-3) \times (-4) = +12$. Then $10 - 12 = -2$. Adding the products gives $+22$; a sign slip gives $+2$.
- 45. (B)** $2 - 5 = -3$; $-3 - 4 = -7$; $-7 + 3 = -4$ °C. Treating the rise as a fall gives -14 ; a sign slip gives $+4$; ignoring the rise gives -7 .
- 46. (C)** $(-2) \times (-3) \times (-4) = -24$ (three negatives $\rightarrow$ negative). $(-5) \times (-5) = +25$; $(-6) + (-6) = -12$; $(-10) - (-1) = -9$. The most negative is -24 .
- 47. (A)** Left to right division: $(-64) \div (-8) = +8$; $+8 \div (-2) = -4$. Then $(-4) - (-3) = -4 + 3 = -1$. Treating $-(-3)$ as -3 gives -7 ; sign slips give ± 7 or $+1$.
- 48. (D)** $102 = 100 + 2$, so $(-8) \times 102 = (-8) \times 100 + (-8) \times 2 = -800 + (-16) = -816$. The other splits mishandle the sign of $(-8) \times 2$, which must be -16 , not $+16$, giving the incorrect -784 .
- 49. (B)** Other = $48 \div (-6) = -8$ (positive product with a negative factor needs a negative partner). Check: $(-6) \times (-8) = +48$. Choosing $+8$ gives -48 ; multiplying instead of dividing gives ± 288 .
- 50. (B)** Net = $-30 + 120 - 45 + 80 = +125$. Twice: $2 \times 125 = +250$. Forgetting to double gives 125 ; a sign error on the net gives -250 ; summing magnitudes gives 550 .
- 51. (A)** Descending means each next number is smaller. $6 > 1 > 0 > -4 > -9$ holds. The first distractor is ascending; the others are out of order (1 is not less than 0 , and -4 is not less than -9).
- 52. (C)** Bracket: $(-2) - (-5) = -2 + 5 = 3$. First term: $(-6) \times 3 = -18$. Second: $(-6) \times (-2) = +12$, with the minus in front: $-18 - 12 = -30$. Treating the second product as -12 (then subtracting) flips the result toward -6 ; sign slips give $+30$.
- 53. (A)** Adding, subtracting, or multiplying two integers always gives an integer, so the integers are closed under those three. Division can give a non-integer (e.g. $7 \div 2$), so closure fails for division — any option that includes division is wrong.
- 54. (D)** $(-2)^3 = -8$; $(-3)^2 = +9$; $(-1)^5 = -1$, and minus that is $+1$. Sum: $-8 + 9 + 1 = +2$. Treating $(-1)^5$ as $+1$ (then subtracting) gives 0 ; sign slips on the cube give other values.
- 55. (D)** Need sum -1 , product -30 . Try 5 and -6 : sum = -1 , product = -30 ✓. $(-5) + 6 = +1$ (wrong sum); $(-10) + 3 = -7$; $(-15) + 2 = -13$. Only 5 and -6 satisfy both.
- 56. (C)** Multiply: $5 \times (-4) = -20$; before it is a minus: $-(-20) = +20$. Divide: $(-12) \div 6 = -2$. So $(-9) + 20 + (-2) = 9$. Treating $-5 \times (-4)$ as -20 gives -31 -type errors; ignoring the division gives $+11$.

57. (C) Net move = $-7 + 3 - 5 + 6 = -3$. From +2: $2 - 3 = -1$. A sign slip on one of the moves gives +1 or -3; ignoring a down move gives +5.

58. (C) Rewrite subtractions of negatives as additions: $-14 + 6 + 9 - 4 = (-14 - 4) + (6 + 9) = -18 + 15 = -3$. Treating every ' $- (-)$ ' as a plain subtraction gives -33; a sign slip gives +3.

59. (A) Each negative factor flips the sign; an odd count of flips leaves the product negative. It is never zero (all factors are non-zero), not positive (that needs an even count), and not equal to the count.

60. (D) Top: $(-5) \times (-2) = +10$, then $10 - 4 = 6$. Bottom: $(-1) + (-2) = -3$. So $6 \div (-3) = -2$. Treating the product as -10 changes the numerator; a sign slip on the quotient gives +2.